

An irreducible CAR representation has no nonzero compact image

MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image

theorem

Simplicity and infinite dimensionality rule out compact operators coming from nonzero CAR elements.

Statement

Let A be the completed CAR algebra: the norm completion of the increasing union of $A_n = M_{2^n}(\mathbb{C})$ under the unital embeddings $a \mapsto a \otimes I_2$ (with the fixed coordinate reindexing). Write $\iota_n : A_n \rightarrow A$ for the canonical isometric inclusion and $e_{ij}^{(n)} = \iota_n(E_{ij})$ for its matrix units. A representation is a unital complex star homomorphism into the bounded operators on a complex Hilbert space. Irreducibility means nonzero action and no proper nonzero closed reducing subspace. Let $\pi : A \rightarrow B(H)$ be irreducible on a complex Hilbert space H . If $a \in A$ and $\pi(a)$ is compact, then $a = 0$. In particular $\pi(A) \cap \mathcal{K}(H) = \{0\}$.

Assumptions

The representation is unital and irreducible in the nonzero, closed-reducing-subspace sense. The space H is complete. Compactness is assumed only for the represented operator $\pi(a)$.

Conclusion

No nonzero element of A can act compactly in this representation. Faithfulness makes the vanishing conclusion a statement about a itself, not merely about its image.

This does not say that $B(H)$ has no compact operators. It describes the intersection of its compact ideal with the represented CAR algebra. Infinite dimensionality of A is a proved property of the fixed CAR limit, not an extra assumption on an arbitrary algebra.

Proof route

Simplicity makes the nonzero unital representation faithful. The preimage of the compact operators is a closed two-sided ideal of A . If it were all of A , the identity on H would be compact; this would make H , and then the faithfully represented algebra A , finite-dimensional. That contradicts infinite dimensionality of the CAR limit.

▼ Proof steps

1. The exact result `representation_injective` gives injectivity of π from simplicity and the nonzero irreducible action.
2. Let $J = \{x \in A : \pi(x) \text{ is compact}\}$. Compact operators form a norm-closed two-sided ideal in $B(H)$, and the representation is continuous, so J is a norm-closed two-sided ideal in A .

3. Simplicity yields $J = \{0\}$ or $J = A$. In the second case $\pi(1) = I_H$ is compact, so H is finite-dimensional. Injectivity then embeds A linearly in the finite-dimensional space $B(H)$, a contradiction. Therefore $J = \{0\}$ and $a = 0$.

Main citations

- The stated existence or structural result — `MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image`
- Faithfulness of an irreducible CAR representation — `MathlibAnnex.CStarAlgebra.CAR.representation_injective`
- Simplicity of the CAR limit — `MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_limit`
- Infinite dimensionality of the CAR limit — `MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional`
- Closed compact-preimage ideal and finite-dimensional contradiction — `MathlibAnnex.CStarAlgebra.eq_zero_of_isCompactOperator_of_injective`
- Representation and nonzero irreducibility conventions — `MathlibAnnex.Analysis.CStarAlgebra.Representation.IsIrreducible`

Lean source signature (exact)

```
theorem eq_zero_of_isCompactOperator_image
  (rho : Representation Limit H) (hrho : rho.IsIrreducible)
  {a : Limit} (ha : IsCompactOperator (rho a)) : a = 0
```

In the source	Mathematical meaning
<code>H; Representation Limit H</code>	A complete complex Hilbert space H and a unital complex-linear star homomorphism $\pi : A \rightarrow B(H)$, with A the completed CAR algebra. Its value at a is a bounded complex-linear operator on H .
<code>rho; hrho : rho.IsIrreducible</code>	The representation is named π in the text. Assume its action is nonzero and has no proper nonzero closed reducing subspace.
<code>{a : Limit}</code>	An arbitrary element $a \in A$, inferred if possible from the compactness hypothesis; braces do not impose an additional condition on a .
<code>ha : IsCompactOperator (rho a)</code>	The operator $\pi(a) : H \rightarrow H$ is compact: the image of the closed unit ball has compact closure. Compactness is a property of its operator image, not of the element a as a map.
<code>a = 0</code>	The conclusion is that a itself is zero in A , not just that $\pi(a)$ is the zero operator.

Further source notes: The source concludes `a = 0`; it does not only conclude that `rho a` vanishes.

Earlier published Card and PDF

▼ **Exact content identity**

Declaration: `MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image`

Accepted content SHA-256: 00fd316c67ed3da3aa765b70633ce2e24039a792d2d14401f89ba81c6f781460

Accepted source guide SHA-256: d13fb3e33c31eaaa8e29c6302e742e4b39b2390883c7c3d82240697ce08bfe6a

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73